



# East West University

CSE 475

Machine Learning

Task – 01

Group – 10

**Submitted To:**

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## Dataset Summary

- **Dataset Title:** putEMG 24-channel hand-gesture dataset
- **Source:** Poznan University of Technology — Nextcloud public share
- **Domain / Application:** Surface-EMG hand-gesture recognition
- **Input Data:** 264 trial ZIP files containing 394,589,696 raw signal samples across 44 distinct subjects
- **Processed Dataset:** 619,479 window samples derived using a sliding window strategy (1,024 samples per window with 50% overlap / 512-sample step size)
- **Features:** 192 features extracted per window (8 time-domain metrics  $\times$  24sEMG channels)
  - Metrics: Mean Absolute Value (MAV), Root Mean Square (RMS), Standard Deviation (STD), Variance (VAR), Waveform Length (WL), Zero Crossings (ZC), Slope Sign Changes (SSC), and Integrated EMG (IEMG)
- **Classes:** 8 gesture classes — 0: Idle, 1: Fist, 2: Flexion, 3: Extension, 6: Pinch-Index, 7: Pinch-Middle, 8: Pinch-Ring, 9: Pinch-Small.

## Data Quality & Integrity

- **Missing Values:** 0 missing feature values (0.00%). No imputation is required prior to modeling.
- **Exact Duplicates:** 0 exact duplicate feature rows.
- **Near-Constant Features:** None found; all 192 features exhibit variance across the dataset.
- **Outliers:**
  - Using the Interquartile Range (IQR) rule ( $Q_1 - 1.5 \cdot \text{IQR}$  or  $Q_3 + 1.5 \cdot \text{IQR}$ ), 8,768,426 feature cells across 223,817 window rows (36.13%) were flagged.

- *Note:* These reflect natural high-amplitude sEMG muscle activity during strong contractions rather than measurement errors. Any scaling or clipping must be fitted strictly inside training folds.
- **Multicollinearity:** 209 feature pairs exhibit high linear correlation ( $|r| > 0.90$ ). Feature selection or dimensionality reduction (e.g., PCA) may help optimize training without information loss.

## Class Distribution & Imbalance

### Class Breakdown:

- 0: Idle — 380,337 windows (61.40%)
- 1: Fist — 35,638 windows (5.75%)
- 2: Flexion — 35,356 windows (5.71%)
- 9: Pinch-Small — 33,799 windows (5.46%)
- 6: Pinch-Index — 33,649 windows (5.43%)
- 7: Pinch-Middle — 33,552 windows (5.42%)
- 3: Extension — 34,228 windows (5.53%)
- 8: Pinch-Ring — 32,920 windows (5.31%)

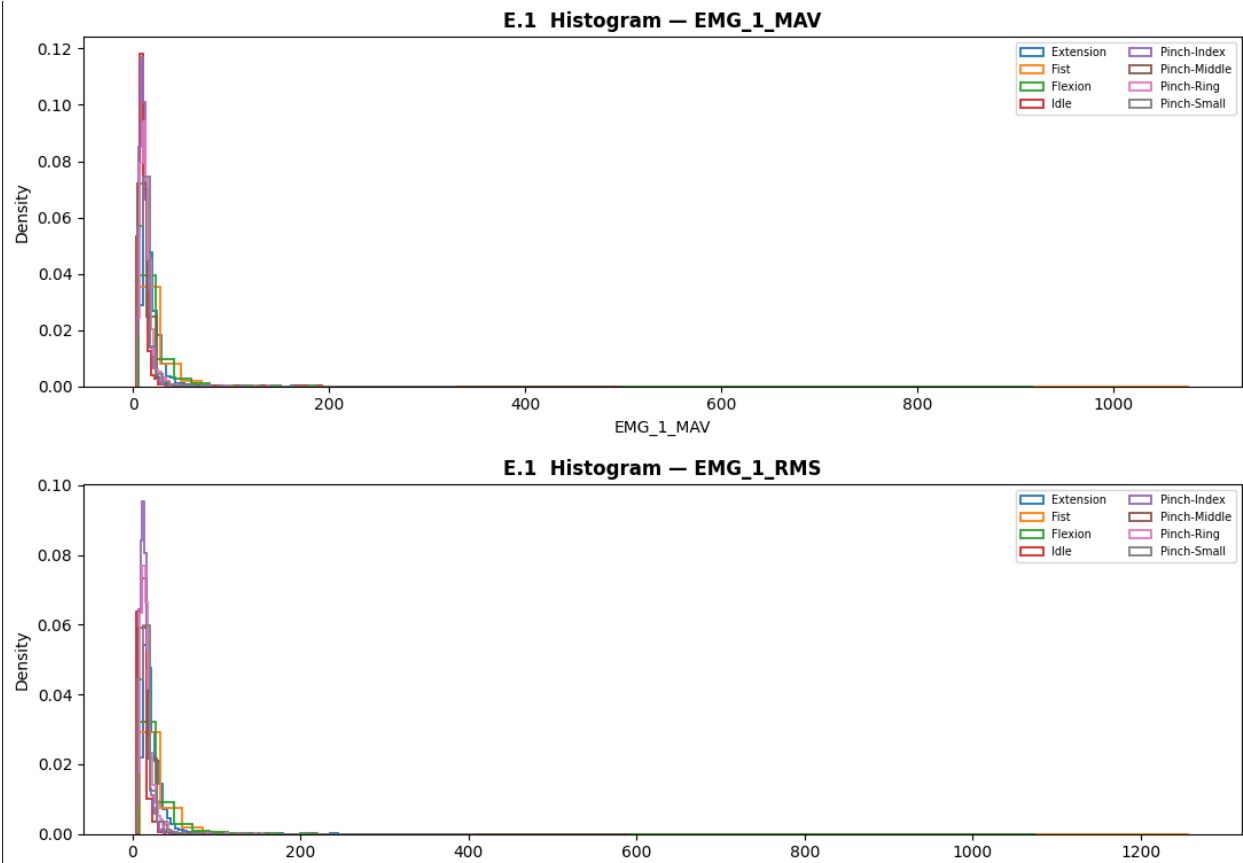
**Imbalance Ratio: 11.55 : 1** (Substantial imbalance due to the majority Idle state).

**Majority-Class Baseline:** A naive classifier predicting Idle achieves a baseline accuracy of 61.40%.

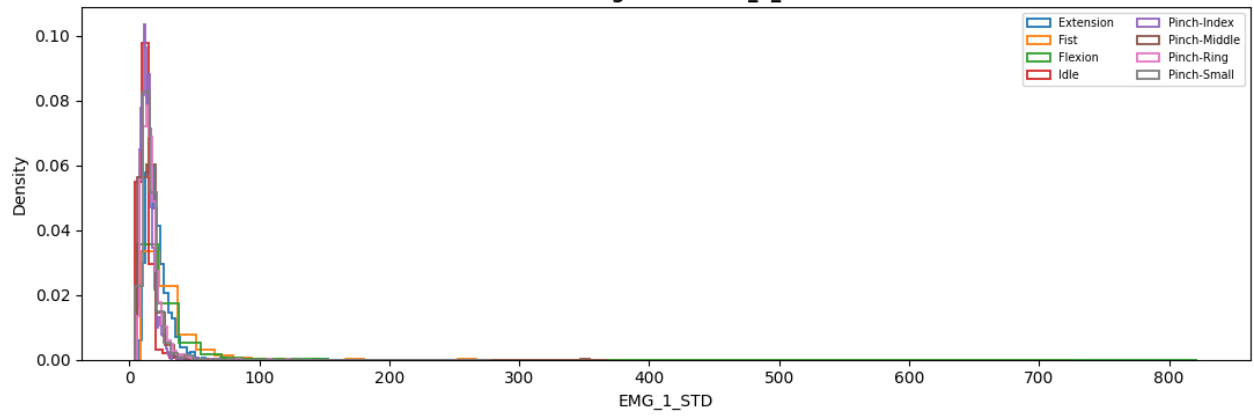
### Modeling Recommendations:

- Evaluate models using **Macro-F1**, **Balanced Accuracy**, per-class precision/recall, and confusion matrices rather than standard accuracy alone.
- Keep consecutive, overlapping windows from the same subject/recording grouped together within train/test splits to prevent data leakage.

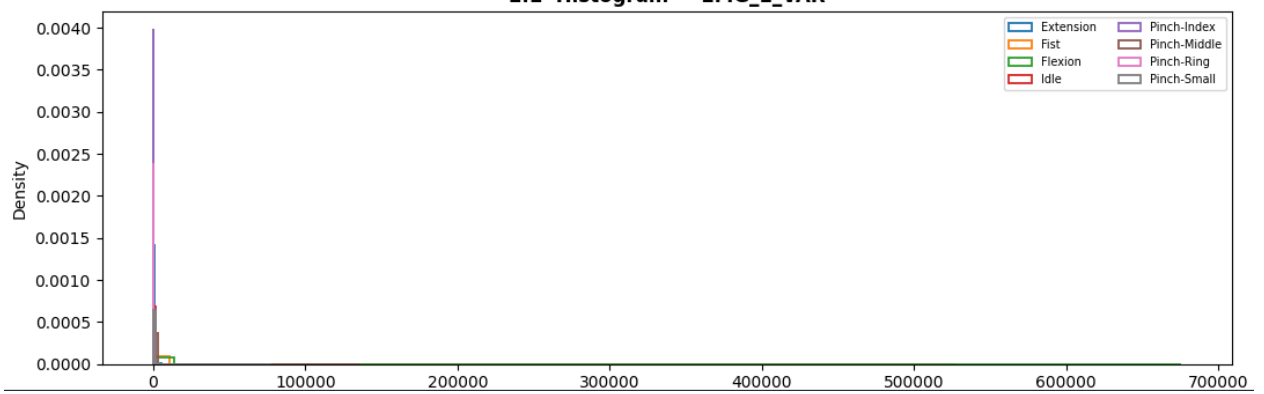
# Histogram



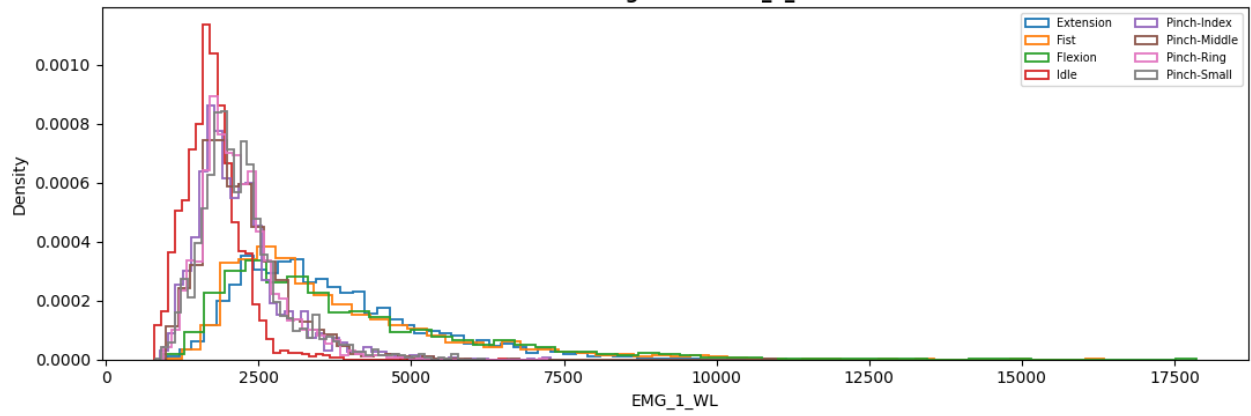
E.1 Histogram — EMG\_1\_STD



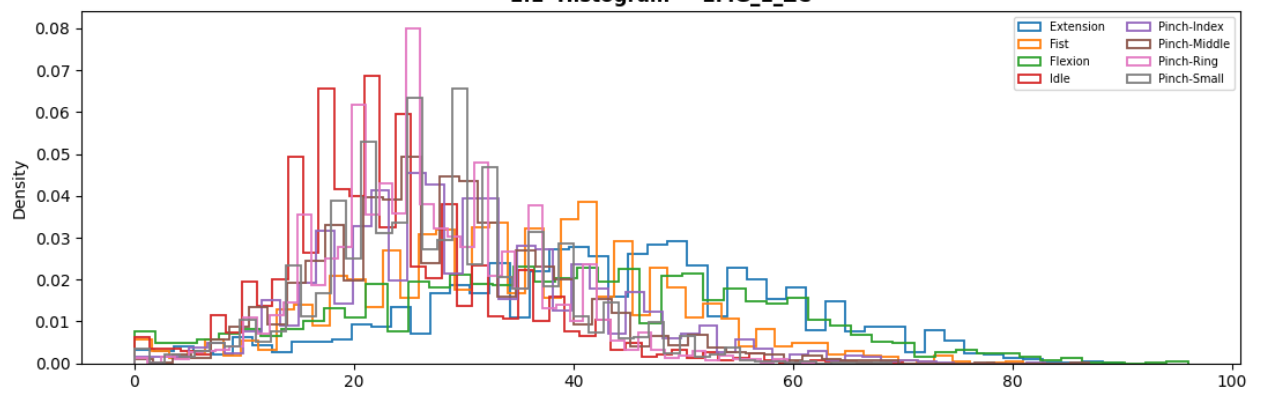
E.1 Histogram — EMG\_1\_VAR

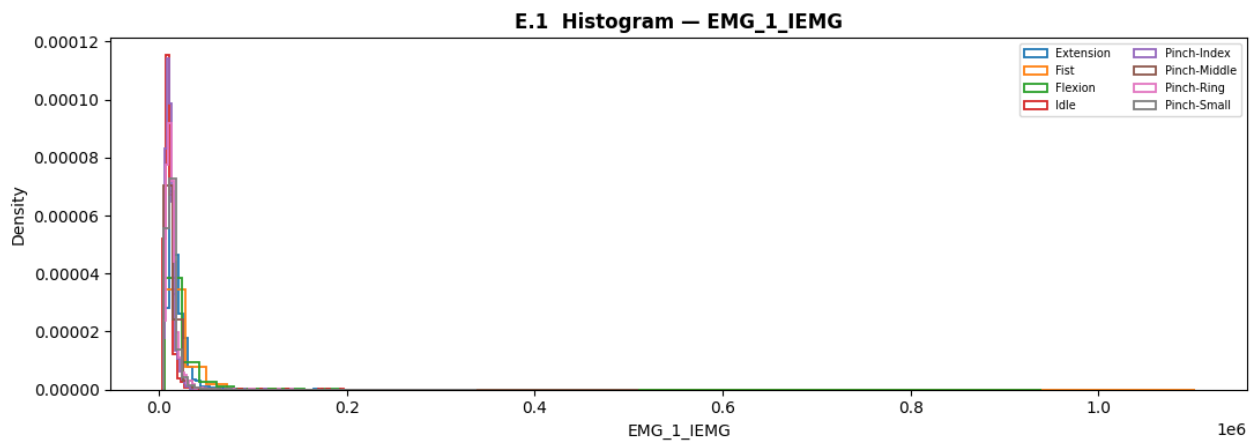
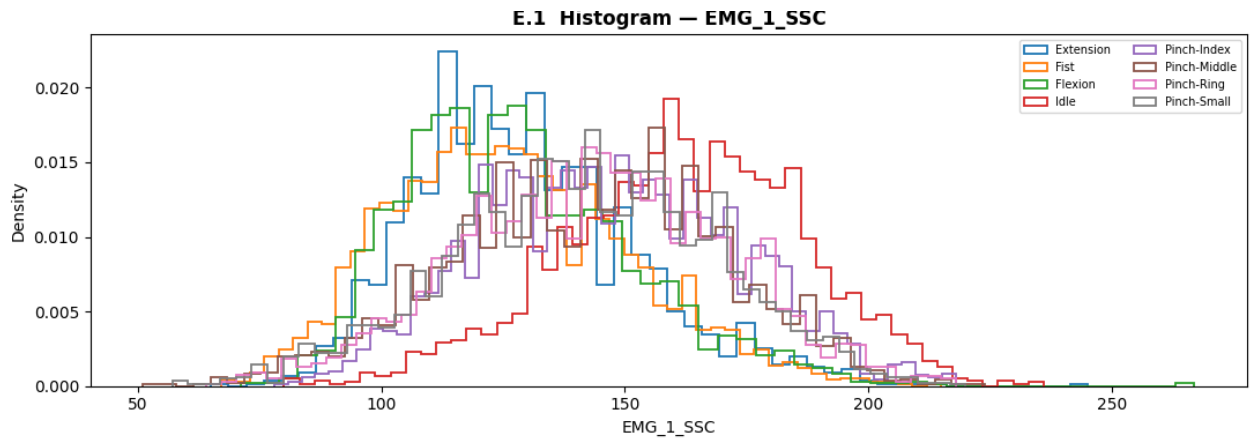


E.1 Histogram — EMG\_1\_WL



E.1 Histogram — EMG\_1\_ZC





## Boxplot

